1 Complex Differentiability

Let $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$ on some region G containing an interior point z_0. If $f(z)$ satisfies the Cauchy–Riemann equations

$$u_x(x, y) = v_y(x, y), \qquad u_y(x, y) = -v_x(x, y),$$

and has continuous partial derivatives in the neighbourhood of z_0, then $f'(z_0)$ exists, where

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

This function is said to be **complex differentiable**, or **analytic**, or **holomorphic**.

If a complex function $f(z)$ is analytic on a region R, it is infinitely differentiable in R.

Definition 1. *Let f_1 and f_2 be analytic functions on Ω_1 and Ω_2 respectively. Suppose $\Omega_1 \cap \Omega_2$ is non-empty and $f_1 = f_2$ on $\Omega_1 \cap \Omega_2$. Then f_2 is called an **analytic continuation** of f_1 to Ω_2.*

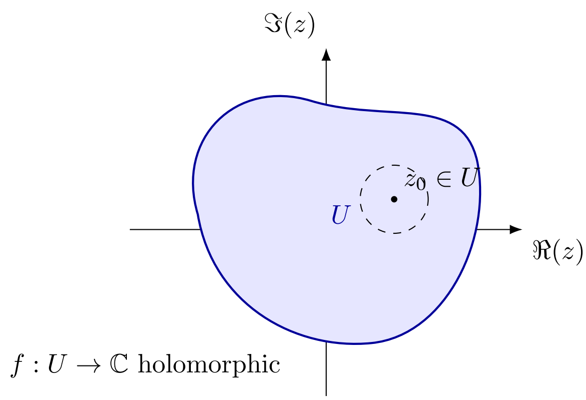

Figure 1: A connected open set $U \subset \mathbb{C}$ (a typical domain on which holomorphic functions are defined).

2 The Riemann Zeta Function

The **Riemann zeta function** is defined over the complex plane as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad s \in \mathbb{C}, \quad \mathrm{Re}(s) > 1.$$

Let $s = \sigma + it$ be the complex variable used. Through analytic continuation, $\zeta(s)$ extends to a meromorphic function on all of $\mathbb{C}$ with a sole singularity being a simple pole at $s = 1$. The extended function satisfies the Riemann functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\, \zeta(1-s)$$

where Γ denotes the gamma function.

3 Trivial Zeros

- For $\mathrm{Re}(s) > 1$, the Dirichlet series $\zeta(s) = \sum n^{-s}$ converges absolutely and is manifestly nonzero (in fact it's easy to show via the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, which converges and has no zero factors when $\mathrm{Re}(s) > 1$).
- Now plug $s = -2n$ (negative even integer) into the functional equation. Then $1 - s = 2n + 1 > 1$, so $\zeta(1 - s) = \zeta(2n + 1)$ is finite and nonzero.
- $\Gamma(1 - s) = \Gamma(2n + 1) = (2n)!$, a finite nonzero number.
- The only factor that can vanish is $\sin(\pi s/2) = \sin(-\pi n) = 0$.

So the product is

$$(\text{finite nonzero}) \times \sin(-\pi n) \times (\text{finite nonzero}) = 0.$$

That forces $\zeta(-2n) = 0$ for every positive integer n.

This also explains why there are no other zeros from this mechanism: at $s = 0$, the sine factor $\sin(0) = 0$ too, but $\Gamma(1-s)$ has no pole there and $\zeta(1-s) = \zeta(1)$ is also a pole, and the two singular behaviors cancel which is why $s = 0$ is not a zero, and also produces the infamous result $\zeta(0) = -1/2$. At positive even integers $s = 2n$, $\sin(\pi s/2) = 0$ too, but that's the domain $\mathrm{Re}(s) > 1$ side (or covered by convergent zeta directly), not producing new information the same way. Hence the trivial zeros specifically come from negative even integers where $\zeta(1 - s)$ sits safely in the region of absolute convergence.